Part B: Gradients, Jacobians, Curvature and Backprop

Sessions 5.1–5.4, by hand

Mohsin Raza · Fall 2026 **Due:** Tuesday 10 November 2026, start of class

How to work. By hand, with a calculator. Show every step: an answer without working gets no credit, and a correct method with an arithmetic slip still earns most of the marks. You may check your answers with R or Python afterwards, but the working you submit must be your own. Vectors are columns, as in class.
Submit: hand in your written work at the start of class on Tuesday 10 November. If you can't attend, upload a photo or scan to Google Classroom (Classwork → Module 5) by the same time.

B1. The gradient and its direction [12 marks]

Let $f(x, y) = x^2 + 4y^2$ and stand at the point $P = (2, 1)$.

(a) Compute ∇f and evaluate it at P. [2]

(b) Compute the rate of change of f at P in the direction of $(1, 1)$. *Remember to normalize.* [2]

(c) What is the largest possible rate of change at P, and in which direction does it occur? [2]

(d) Find a unit direction at P in which the instantaneous directional derivative of f is zero. Explain its geometric relationship to the gradient and to the level curve through P. [2]

(e) The minimum of f is at the origin. Find the angle between the steepest-descent direction $-\nabla f(P)$ and the straight line from P to the origin. [2]

(f) From which points does $-\nabla f$ point exactly at the minimum? Explain briefly. [2]

B2. Jacobian and chain rule through a layer [15 marks]

A tiny layer takes $\mathbf{x} \in \mathbb{R}^2$, computes $\mathbf{z} = W\mathbf{x}$, then applies ReLU elementwise: $\mathbf{a} = \mathrm{ReLU}(\mathbf{z})$. The loss compares $\mathbf{a}$ with a target $\mathbf{t}$:

$$W = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{t} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad L = \tfrac{1}{2}\|\mathbf{a} - \mathbf{t}\|^2.$$

(a) Forward pass: compute $\mathbf{z}$, $\mathbf{a}$ and L. [3]

(b) Write down the Jacobian of $\mathbf{z}$ with respect to $\mathbf{x}$, and the Jacobian of $\mathbf{a}$ with respect to $\mathbf{z}$ at this point. [3]

(c) Use the chain rule to get the Jacobian of $\mathbf{a}$ with respect to $\mathbf{x}$. State its shape. [3]

(d) The gradient of L with respect to $\mathbf{a}$ is $\mathbf{a} - \mathbf{t}$. Use the chain rule to compute $\nabla_{\mathbf{x}} L$. *Hint: with column gradients,* $\nabla_{\mathbf{x}} L = J^\top \nabla_{\mathbf{a}} L$. [4]

(e) One row of your Jacobian in (c) is all zeros. Explain why. What does this imply about gradient propagation through that ReLU unit at this input? Does it mean the entire layer cannot learn? Explain briefly. [2]

B3. Critical points and the Hessian [18 marks]

Let $g(x, y) = x^3 + y^3 - 3xy$.

(a) Find all critical points of g (where $\nabla g = \mathbf{0}$). *There are two.* [4]

(b) Compute the Hessian $H(x, y)$. [3]

(c) Classify each critical point using the eigenvalues of H at that point. [5]

(d) At the critical point that is a minimum, use the second-order Taylor approximation to estimate g at a step $\boldsymbol{\Delta} = (0.1, 0.1)$ away. Compare with the exact value. [3]

(e) At that minimum, find a unit eigenvector of H for its largest eigenvalue, and compute the 2-norm condition number $\kappa_2(H)$. Along which direction is the local curvature greatest? [3]

B4. Backpropagation by hand [20 marks]

A one-input network with one hidden ReLU unit and biases:

$$z = w_1 x + b_1, \qquad h = \mathrm{ReLU}(z), \qquad \hat{y} = w_2 h + b_2, \qquad L = (\hat{y} - y)^2,$$

with $x = 3$, $y = 2$, $w_1 = 0.5$, $b_1 = -1$, $w_2 = 2$, $b_2 = 0.5$.

(a) Forward pass: compute z, h, $\hat{y}$ and L, in a table. [4]

(b) Backward pass: compute $\partial L/\partial \hat{y}$, $\partial L/\partial w_2$, $\partial L/\partial b_2$, $\partial L/\partial h$, $\partial L/\partial z$, $\partial L/\partial w_1$ and $\partial L/\partial b_1$ in a table. For each one, show which upstream gradient it uses and the local derivative it is multiplied by. Reuse intermediate gradients where appropriate. [8]

(c) Gradient check for w_1: compute L at $w_1 = 0.51$ and at $w_1 = 0.49$ (everything else unchanged). Give the forward-difference estimate $\big(L(0.51) - L(0.5)\big)/0.01$ and the central-difference estimate $\big(L(0.51) - L(0.49)\big)/0.02$. Which is closer to your answer from (b), and why? [5]

(d) Now change b_1 to -2. Without redoing the whole backward pass, what are $\partial L/\partial w_1$ and $\partial L/\partial b_1$, and why? [3]

B5. Gradient flow [10 marks]

In a deep network, the gradient reaching the first layer is (roughly) a product of one factor per layer. In (a) and (b), consider only the activation derivatives, ignoring the effect of the weight matrices.

(a) Each layer uses a sigmoid at its steepest point. What factor does each activation contribute, and what is the product over 10 layers? [3]

(b) Repeat for tanh at its steepest point, and for an active ReLU. [2]

(c) Suppose instead each layer multiplies the gradient by about 1.5. What happens over 20 layers? What would you expect to see during training? [3]

(d) Name one design choice from Session 5.4 that addresses (a), and one that addresses (c). [2]

Reading: *Mathematics for Machine Learning*, Sections 5.2, 5.3, 5.6, 5.7 and 5.8.